# Response to the committee

*A Lightweight Canonicalization Engine in C for Disciplined Convex Programming*

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To Prof. Sébastien Le Digabel (chair), Prof. Antoine Legrain (second member) and Prof. Antoine Lesage-Landry (supervisor).

## Message au jury

Bonjour chers membres du jury,

Je voulais tout d'abord vous remercier pour vos commentaires sur mon mémoire. Comme vous m'avez demandé de clarifier et d'approfondir plusieurs points des résultats, j'ai pris le temps de les rendre plus complets, et j'ai aussi découvert quelques erreurs dans les affirmations que j'avais faites sur les contributions.

Premièrement, il y a maintenant quatre problèmes qui sont plus lents. J'explique la cause de chacun au chapitre 4 (§4.1), avec la part de chaque phase du temps (tableau 4.2), et je donne le détail à la section 3. Pour deux d'entre eux, l'écart disparaît presque ou s'inverse avec des changements dans le « frontend » (par frontend, je fais référence à la manipulation des arbres d'expression avant le calcul des dérivées) : UnconstrainedQP passe à 16 ms contre 13 ms pour `CVXPY`, avec le même changement pour tous les backends, et OptimalAdvertising devient plus rapide (0,33 s contre 0,53 s). Par exemple, pouvoir détecter que `D @ ones` (où `D` est une matrice quelconque) équivaut à `D.sum(axis=1)`, la somme sur les colonnes, est un grand avantage. Un troisième, ConvexPlasticity, devient 2,5 fois plus rapide avec un test de densité, mais reste plus lent que `CVXPY`. Ces changements ne règlent pas tout : une partie du ralentissement vient de `SparseDiffEngine` lui-même (voir Q5).

Deuxièmement, j'ai refait toutes les mesures sur un ordinateur plus récent (MacBook M2 de 2022, avec Apple `Accelerate`), plutôt que sur l'ordinateur Intel de 2018 de la version soumise, pour que les résultats soient reproductibles sur du matériel actuel, comme celui de la plupart des utilisateurs de `CVXPY`. Pour la première canonicalisation, `diffengine` reste le plus rapide sur 21 des 25 problèmes, mais l'accélération moyenne passe de 2,8 à 2,3 fois, en partie à cause d'UnconstrainedQP (troisième point). Pour les re-résolutions paramétriques, en revanche, `diffengine` n'est plus rapide que sur trois des sept problèmes, contre cinq dans la version soumise ; sur les quatre autres, la plus grande partie d'une re-résolution est du post-traitement, après `SparseDiffEngine`. J'ai donc modifié ma conclusion et la discussion des résultats pour dire seulement qu'il y a des bénéfices, mais que la séparation ne remplace pas encore le tenseur pour les re-résolutions. Réduire ce post-traitement est maintenant le premier des travaux futurs.

Troisièmement, le problème UnconstrainedQP n'était pas réellement 100 fois plus rapide.

C'était dû à un changement de frontend que j'avais appliqué sur la branche du backend `diffengine`, mais pas sur celles des autres. En l'appliquant à tous, cet avantage disparaît. Ceci a été rectifié : aucun backend n'utilise maintenant ce changement.

À part ces trois points, qui sont les changements majeurs, j'ai aussi fait les études d'ablation et exécuté plus d'instances de benchmark, avec des mesures de signification statistique, ce qui répond à tous vos commentaires.

# 1 Summary of corrections

**Main findings.** The new measurements answer the committee's comments as follows.

- The dense blocks help only on problems with dense data. They slow down the three problems that build a dense block from data that is mostly zeros, and change little on the others.
- Over the suite, the speedup remains almost the same without the dense blocks.
- Without the symbolic and numeric split, a re-solve repeats the conversion and the symbolic pass and is 2.9 times slower.
- The speedup holds across sizes, except on three of the slower problems. Where `diffengine` runs out of memory and `CVXPY` does not, the dense blocks are usually the cause.
- The dense blocks pay off only when about half or more of the entries are nonzero.
- Each of the four slower problems has an identified cause, and three of them improve with a change made before the engine.

I have made all the modifications required by the committee and by Prof. Antoine Legrain's examiner report, as well as the minor corrections that Prof. Sébastien Le Digabel annotated and those listed in the appendix of Prof. Legrain's report. Prof. Le Digabel's corrections are not listed here; those of Prof. Legrain are in Table 2. All changes since the version sent to the committee are marked in `Document-diff.pdf`, with added text in blue and deleted text in red. Section, table and page numbers below refer to the final version. The code of the benchmarks of Chapter 4 of the thesis and of Section 3 below is in the repository `https://github.com/SparseDifferentiation/canonicalization-benchmarks`.

I wrote first drafts of these answers myself and used Claude, the AI tool declared in the thesis, to help with the results and the writing.

Table 1 lists the required modifications. "J" refers to the committee's requirements after the defence, and (i)–(vii) to the items of the synthesis of the examiner report. The defence questions are answered in Section 2; the other items are in the thesis sections listed.

Table 1: Required modifications and where they are made.

| Request | Change | Where |
|---|---|---|
| J1a: statistical significance | 30 runs per comparison; Mann–Whitney [1] per problem, Holm-adjusted [2]; Wilcoxon [3] over the suite. | Ch. 4 introduction, p. 51; §4.1 |
| J1b, (ii): ablation | Dense blocks turned off on all 25 problems, in time and memory; the split ablated on the re-solves (Q2). | §4.1, p. 57, Tables 4.3 and 4.4 |
| J1c: more instances | New parameter values for each run, every problem scaled in size, and eight in density. | Ch. 4 introduction; §4.2 |
| J1d: sensitivity to size and sparsity | All 25 problems at five sizes and eight at seven densities, with and without the dense blocks. | §4.2, p. 58, Tables 4.5 and 4.6 |
| J1e, (iii): non-DPP problems | Re-solves of two families in DPP, non-DPP and parameter-free variants, and of ConvexPlasticity (Q3). | §4.3, p. 60 |
| J2, (i): related literature and DNLP | Matrix-free modelling, `CVXcanon`, `CVXGEN`, `Convex.jl` 0.16, `Pyomo`, block-sparse antecedents and sparse factorization; the authorship of `SparseDiffEngine` and what is specific to this work (Q6). | §2.2, §1.5, §3.1.1 |
| J3: declaration of AI use | New front-matter page. | Front matter |
| (iv): repeated measurements | 30 runs, each in a fresh process; tables give mean ± standard deviation. | Ch. 4 introduction; Table 4.1 (here Table 7) |
| (v): dimensions of the instances | Dimensions, nonzeros, density, parameter size and DPP status of every problem. | Appendix A, Tables A.1 and A.2 (here Tables 14 and 15) |
| (vi): time by phase | The share of each phase on the four slower problems, with the cause of each. | §4.1, p. 52, Table 4.2 |
| (vii): correctness check | $\mathbf{P}$, $\mathbf{A}$, $\mathbf{c}$ and $\mathbf{b}$ are checked against the `CPP` backend to within $10^{-9}$ of their largest entry, at the first canonicalization and after a parameter update. | Ch. 4 introduction |
| Point 6: complexity | Storage of a permuted dense block against CSR, and the cost of the symbolic and numeric passes of a product against Gustavson's algorithm. | §3.3.1 |

The text added to the thesis for the literature, the statistics, the correctness check and the complexity is quoted below, with citations omitted.

Changes in the thesis, J2, (i): §2.1.3, `CVXcanon` and `CVXGEN`

the C++ backend `cvxcore` keeps one sparse matrix per parameter slice and loops over the slices at every operator. It began as `CVXcanon`, a C++ library written to replace the Python

matrix stuffing of early versions of `CVXPY`, and was renamed `cvxcore` in 2018. [...] `CVXPYgen` follows the earlier code generator `CVXGEN`, which generates a custom C solver for a family of small quadratic programs.

Changes in the thesis, J2, (i): §2.3.1, `Pyomo`

`Pyomo`, which also writes `.nl` files, avoids a new file per parameter change for solvers with an in-memory interface: its persistent solver interfaces keep the model inside the solver and update the changed parameters and constraints in place.

Changes in the thesis, J2, (i): §2.3.2, `Convex.jl`

`Convex.jl`, the DCP language of the Julia ecosystem, also targets MOI. Since version 0.16, it represents every affine expression as a sparse matrix and an offset over the solver variables, accumulated as the problem is converted to MOI. Like the backends of `CVXPY`, it stores the coefficients in a general-purpose sparse format.

Changes in the thesis, J2, (i): §3.2, block-sparse antecedents

Keeping dense blocks inside a sparse matrix is not a new idea. Finite-element codes assemble dense element matrices into a sparse global matrix, and nonlinear least-squares libraries such as `Ceres` and `g2o` store the block-sparse Jacobians of bundle adjustment as dense blocks. In those codes, the position of each block follows from the structure of the problem, one element or one residual at a time. In our representations, the symbolic pass derives it from the expression tree.

Changes in the thesis, J2, (i): §1.1, the symbolic and numeric split

This separation between structure and values goes back to sparse direct solvers, which analyse the sparsity pattern of a matrix once and repeat the numeric factorization for new values, and to Gustavson's sparse product.

Changes in the thesis, J1a, J1c, (iv): Chapter 4, Statistics

Each measurement is made in 30 runs, each in a fresh process, and on a parametric problem each run uses new parameter values. The tables report the mean and the standard deviation of the runs, and the ratios are ratios of medians. On each problem, the runs of `diffengine` and of the other backend are compared with a two-sided Mann–Whitney $U$ test, with the $p$-values adjusted over the problems by the Holm–Bonferroni method. Over the suite, we test whether the geometric mean of the ratios differs from 1 with a Wilcoxon signed-rank test on their logarithms. [...] Every ratio in the tables of this chapter is significant unless its caption says otherwise.

Changes in the thesis, (vii): Chapter 4, Setup

Before any timing, we check that `diffengine` returns the same problem data as `CPP`, to within $10^{-9}$ of the largest entry, at the first canonicalization and after a parameter update. [Footnote:] Except for one NaN entry that comes from the data of OptimalAdvertising itself.

Changes in the thesis, Point 6: §3.3.1, complexity

A permuted dense matrix with an $m_0 \times n_0$ block stores $m_0 n_0$ values and $m_0 + n_0$ indices, against $m_0 n_0$ values and $m_0 n_0 + m + 1$ indices for the same block in CSR form, where $m$ is the number of rows of the whole matrix. For (3.2), [...] the symbolic pass costs $O(m_A + m_B)$ and the numeric pass performs $n_B \, s \, n_A$ multiply-adds in one `dgemm`, whereas Gustavson's algorithm performs the same multiply-adds after a symbolic pass of $O(n_B \, s \, n_A)$.

Table 2: Minor corrections of Prof. Legrain's report.

| Correction | Change |
|---|---|
| Résumé: « librairie » | « bibliothèque logicielle ». |
| §4.3: « 24 of the 25 » contradicts « 21 of 25 » | « canonicalizes faster than `cvxcore` on 23 of the 25 problems » (§4.4); the count changed with the new measurements. |
| Harmonize the main contribution (§1.5, Ch. 3, §3.5) | The same order everywhere, the matrix classes first and the split second, with no "main contribution". |
| Symbolic/numeric split: cite sparse factorization first | Davis and Gustavson cited before the NLP interfaces (quoted above). |
| Figure 3.5 relies on colour alone | Each cell and the legend carry a letter (a)–(d) (Figure 3.5). |
| References: « p. » for page ranges | Ranges now print as « pp. ». |
| References to Eigen, CppAD and MOSEK incomplete | URLs added; Eigen version 3.4.0; MOSEK documentation. |
| Status of the merged pull request | Merged on 9 September 2026, opt-in, parameter-free problems only; the parametric path is under review (§5.2). |
| The author contributed to `SCIPY` and `COO` | Stated in §5.2. |
| Delineation from DNLP (§1.5, §3.1) | Q6. |
| Table 2.2 and Pyomo | A note under the table and the paragraph quoted above. |

# 2 Replies to the defence questions

**Q1.** *Why materialize dense blocks rather than leave them as operators, as in the matrix-free approach of Diamond and Boyd [4]? When is each approach preferable?*

Because the solvers that `CVXPY` calls need the matrices. Its default solvers are `Clarabel`, an interior-point method, and `OSQP` and `SCS`, which are first-order methods; commercial solvers such as `MOSEK` and `Gurobi` are interior-point or simplex methods. All of them factorize a matrix built from $\mathbf{A}$ and $\mathbf{P}$: an interior-point method factorizes a KKT system at every iteration, and `OSQP` and `SCS`, in their default direct mode, factorize their KKT matrix and reuse the factorization across iterations, refactorizing only when a step-size parameter changes. So

every one of them needs $\mathbf{A}$ and $\mathbf{P}$ explicitly, whatever its algorithm. Matrix-free modelling pays off for very large problems built from fast transforms, solved by first-order methods that only need products with $\mathbf{A}$ and $\mathbf{A}^\top$, such as the indirect mode of `SCS`. A permuted dense block lies between the two: it is a materialized matrix, but it keeps the structure of the operator, namely its row and column selectors and its dense block.

Changes in the thesis, §2.2, p. 25

In convex optimization, Diamond and Boyd take the matrix-free view. Instead of forming $\mathbf{A}$, they represent the problem as a graph of linear operators, available only through their forward and adjoint products, which suits first-order solvers. The solvers that `CVXPY` calls by default instead factorize a matrix built from $\mathbf{A}$ and $\mathbf{P}$. The matrix-free approach pays off on very large problems built from fast transforms, such as convolutions and Fourier transforms, and it needs solvers that accept operators instead of matrices. The permuted dense matrices of §3.2 sit between the two: they form the matrix that the solver factorizes, but keep the structure of the operator, its row and column selectors and its dense block.

**Q2.** *If the permuted dense representations are disabled, what fraction of the speedup remains?*

Most of it. A switch of our `CVXPY` branch stores every constant matrix in the sparse format. With it, `diffengine` is 2.15 times faster than the best `CVXPY` backend over the suite, against 2.29 times with the dense blocks, and the difference is not significant ($p = 0.07$). Per problem, the dense blocks make `diffengine` 1.2 to 3 times faster on twelve problems, and 1.4 to 46 times slower on CVaR, ConvexPlasticity and UnconstrainedQP, which build a dense block from data that is mostly zeros (Table 8). Without them, `diffengine` also uses 0.83 times the peak memory in geometric mean (Table 9). The thesis states these results in §4.1.

The split between the symbolic and numeric passes can be turned off on re-solves only; without it, a re-solve is 2.9 times slower, significantly on all seven problems (Table 3).

Table 3: Not in the thesis. The ablations in summary: geometric mean of the ratio of each variant to its reference, the Wilcoxon $p$-value over the problems, and the number of problems with a significant difference (Mann–Whitney, Holm-adjusted). The two that are not significant, ParametrizedQP on the first canonicalization and SimpleScalarParametrizedLP on the re-solve, have no dense constant data, so turning the dense blocks off barely changes their work.

| Variant | Reference | Ratio | $p$ | Significant |
|---|---|---|---|---|
| *First canonicalization,* 25 *problems* | | | | |
| no dense blocks | `diffengine` | 1.06 | 0.067 | 24/25 |
| `diffengine` | best `CVXPY` | 0.44 | 0.003 | 25/25 |
| no dense blocks | best `CVXPY` | 0.46 | $3.2 \times 10^{-5}$ | 25/25 |
| *Re-solve,* 7 *DPP problems* | | | | |
| no dense blocks | `diffengine` | 1.01 | 0.938 | 6/7 |
| no split | `diffengine` | 2.86 | 0.016 | 7/7 |
| neither | `diffengine` | 3.68 | 0.016 | 7/7 |

**Q3.** *How does re-solving behave on a non-DPP parametric problem, for instance with a product of two parameters or a parameter in a non-affine position?*

On the two families written with a product of two parameters, which DPP forbids, `CVXPY` has to canonicalize the problem again at every solve, while `diffengine` keeps its symbolic pass and reruns only the numeric pass. It re-solves as fast as on the DPP variant, 13 to 14× faster than `CVXPY` on `lasso` and `factor_cov`, significantly on both (Table 4). ConvexPlasticity, the suite's only natural non-DPP parametric problem, breaks DPP for the second reason: its parameters lie inside the argument of a quadratic form, which `CVXPY` requires to be parameter-free for a solver that accepts a quadratic objective. It is the exception: it re-solves in 207 ms against 42 ms for `CVXPY`, because 88 % of the re-solve assembles a dense 3,000 × 3,000 block of $\mathbf{P}$ built from a diagonal. Without the dense blocks, it re-solves in 2.0 ms, 21× faster than `CVXPY`. The thesis no longer claims that the split replaces the tensor (Q4).

Table 4: Not in the thesis. Re-solve time in milliseconds, mean over 30 runs; every standard deviation is below 1 % of the mean. `factor_cov` is FactorCovarianceModel with $n$ assets. The non-DPP variant writes $\lambda$, or the risk aversion $\gamma$, as the product of two parameters, and the parameter-free variant makes it a constant. `CVXPY` applies the tensor on the DPP variant and canonicalizes again on the others; the cached column of `diffengine` runs the numeric pass only, and the rebuild column the symbolic pass as well. The ratio is the median of `diffengine`, cached when it can, over that of `CVXPY`, * when significant.

| | | | `diffengine` | | |
|---|---|---|---|---|---|
| Family | Variant | `CVXPY` | cached | rebuild | Ratio |
| `lasso`, $n = 1{,}000$ | DPP | 6.4 | 23.3 | 32.7 | 3.64* |
| | non-DPP | 315 | 23.3 | 32.8 | 0.07* |
| | parameter-free | 338 | — | 49.0 | 0.14* |
| `factor_cov`, $n = 5{,}000$ | DPP | 1.2 | 3.6 | 6.9 | 2.86* |
| | non-DPP | 45.8 | 3.6 | 7.0 | 0.08* |
| | parameter-free | 47.1 | — | 11.0 | 0.23* |

The thesis states the results of Table 4 in §4.3 without the table:

Changes in the thesis, §4.3, p. 60

The numeric pass needs a fixed expression tree, not the DPP rules. We write the Lasso problem and FactorCovarianceModel with one parameter as the product of two, which DPP forbids. `CVXPY` then canonicalizes again at every solve, while `diffengine` reruns only its numeric pass, 13 to 14 times faster. The exception is ConvexPlasticity, where the re-solve is spent assembling a dense block of $\mathbf{P}$.

Changes in the thesis, §4.4, p. 61

Caching canonicalization does not necessarily require the DPP ruleset. [...] the symbolic and numeric split re-solves within a factor of four of the cached tensor on every DPP problem. [...] it re-solves problems that are not DPP, on which `CVXPY` has to canonicalize again, 13 to 14

times faster.

**Q4.** *Why not re-evaluate only the subtrees that depend on the updated parameters?*

The engine now does this: after a parameter update, the numeric pass re-evaluates only the subtrees that contain a parameter, and keeps the cached values of the others. All re-solve times in the thesis use it.

The prediction of the examiner report, that this would make `diffengine` at least as fast as the tensor in every case, does not hold. On the seven DPP problems (Table 5), `diffengine` has the fastest first canonicalization on all seven, 2.5 to 6.6× faster than the fastest tensor build. The cached tensor re-solves faster on four of them, by 1.10 to 3.60×, and recovers its slower first canonicalization only after 18 to 124 re-solves. On those four, the numeric pass takes only 4 to 24 % of a re-solve; the rest is post-processing after the engine, the conversion to the solver's format. We conclude that the split does not replace the tensor on DPP problems that are re-solved many times, but comes within a factor of four of it without the DPP rules. Making the post-processing cheaper is now the first item of future work.

Table 5: Table 4.7 of the thesis (p. 60): first canonicalization time in seconds and re-solve time in milliseconds on the seven DPP parametric problems, mean over the runs, with the standard deviation of the re-solves. A dash denotes a tensor that exceeded the memory limit or the time limit. Bold denotes the fastest first canonicalization and the faster re-solve. The tensor column is the fastest of the three `CVXPY` backends, which agree within 5 %. The ratio is the median re-solve time of `diffengine` over that of the tensor. Every ratio is significant.

| | first canonicalization (s) | | | | re-solve (ms) | | |
|---|---|---|---|---|---|---|---|
| Benchmark | `CPP` | `SCIPY` | `COO` | `diffengine` | tensor | `diffengine` | Ratio |
| ParamSmallMatrixStuffing | 2.02 | 7.30 | 0.083 | **0.033** | 1.84 ± 0.01 | **0.438 ± 0.007** | 0.24 |
| ParamConeMatrixStuffing | 0.74 | 1.85 | 0.055 | **0.022** | 0.958 ± 0.005 | **0.265 ± 0.004** | 0.27 |
| SimpleFullyParametrizedLP | — | — | 0.57 | **0.12** | 46.7 ± 0.1 | **34.6 ± 0.8** | 0.73 |
| SimpleScalarParametrizedLP | 1.33 | 1.17 | 1.19 | **0.23** | **62.8 ± 0.3** | 70.4 ± 5.9 | 1.10 |
| ParametrizedQP | — | — | 2.63 | **0.76** | **299 ± 8** | 349 ± 16 | 1.16 |
| SVMWithL1Regularization | 2.80 | 2.57 | 2.83 | **0.39** | **38.3 ± 0.1** | 125 ± 2 | 3.26 |
| FactorCovarianceModel | 1.20 | 1.14 | 1.19 | **0.18** | **20.5 ± 0.1** | 74.2 ± 0.8 | 3.60 |

Changes in the thesis, §3.4.1, p. 47

The numeric pass therefore skips an affine tree after the first evaluation, until the next parameter update. After an update, it re-evaluates only the subtrees that contain a parameter, and keeps the cached values of the others.

**Q5.** *When does the global layout become a handicap relative to `CVXPY`'s local layout, in particular for memory?*

When the problem has many variables and many nodes. In the global layout, each node keeps, besides its Jacobian, integer work arrays of length $n$, the total number of variables, whatever variables its own subtree contains. Their memory is therefore $O(\text{nodes} \times n)$ integers.

In CVXPY's local layout, each node stores blocks only for the variables it depends on, so the same overhead is proportional to the sum, over the nodes, of the number of variables each one depends on. The two coincide when every node depends on every variable, and the local layout wins when most nodes depend on a few variables out of many.

OptimalAdvertising shows it. Each of its 250 minimum terms depends on one row of a $250 \times 1{,}000$ variable, but has a Jacobian over all 250,000 variables. Without the dense blocks, its peak memory is 945 MB, against 173 to 197 MB for the CVXPY backends. With them it reaches 6.1 GB, because each dense block of the aggregate constraints $\mathbf{D}^\top \mathbf{1} \leq \mathbf{t}$ and $\mathbf{D1} \geq \mathbf{c}$ also keeps one integer per variable of the problem: the same cost, paid per block rather than per node. Rewriting these products with a vector of ones as sums avoids the dense blocks: OptimalAdvertising then canonicalizes in 0.33 s instead of 0.91 s, faster than the 0.53 s of CVXPY, with 1.4 GB instead of 6.1 GB.

**Q6.** *What part of* ***SparseDiffEngine*** *pre-existed the thesis or was developed for DNLP, and what part is specific to this work?*

SparseDiffEngine is joint work with Daniel Cederberg. It was first built as the differentiation engine of the DNLP extension of CVXPY [5], which differentiates nonlinear atoms at every iterate of the solver. The version sent to the committee should have said so clearly; it now does. This thesis uses the engine for a different purpose, canonicalization: it evaluates the affine atoms and the quadratic form at the origin to obtain the problem data. What is specific to the thesis is the interface to CVXPY, namely the diffengine backend and its re-solve path for parametric problems, and the benchmarks of Chapter 4.

Changes in the thesis, §1.5, p. 16

SparseDiffEngine is joint work with Daniel Cederberg. It was first built as the differentiation engine of the DNLP extension of CVXPY, where it differentiates nonlinear atoms such as exp, log, and entr at every iterate of the solver. This thesis builds on the observation that canonicalization is a special case of the same computation: once a problem is canonicalized, only affine atoms and quadratic forms remain, and their derivatives are the problem data.

Changes in the thesis, §3.1.1, p. 37

What is specific to this thesis is the interface to CVXPY of §3.4, namely the diffengine backend and its re-solve path for parametric problems, and the benchmarks of Chapter 4.

**Q7.** *What is the strategy for BLAS and binary distribution of* ***sparsediffpy****?*

cvxcore is a C++ extension that lives inside the CVXPY repository, so CVXPY has to compile and ship it. sparsediffpy is a separate package that CVXPY only depends on. Its compiled wheels for Linux, macOS and Windows are published on PyPI; they link Apple's Accelerate on macOS and bundle OpenBLAS elsewhere, so users install nothing beyond the package. The binary does not go away, but it is built and released independently of CVXPY, like its other compiled dependencies, which is what the removal of cvxcore aims at.

# 3 Results: the new tables and what they show

The thesis adds five tables to Chapter 4: the phases, the peak memory, the dense blocks, and the density and size sweeps of §4.2. It also adds the two tables of Appendix A and redoes two tables of Chapter 4 (Table 6). To keep the thesis short, three other tables are not in it: Tables 3 and 4 in the replies above, and the one below. The thesis states their results in a sentence or two, and I give them here in full. After each table I say what I conclude from it. All are measured as in Chapter 4, with the code at `https://github.com/SparseDifferentiation/canonicalization-benchmarks`.

Table 6: Tables added to the thesis, or revised.

| Item | Content | Request |
|---|---|---|
| Table 4.1 | Revised: first canonicalization of the 25 problems, now mean ± standard deviation over 30 runs with significance, and the ratio without the dense blocks, p. 53; here Table 7 | J1a, J1b, (ii), (iv) |
| Table 4.4 | First canonicalization of `diffengine` with and without the dense blocks, grouped by their effect, p. 57; here Table 8 | J1b, (ii) |
| Table 4.2 | Phases of the first canonicalization on the four slower problems, p. 54; here Table 10 | (vi) |
| Table 4.3 | Peak memory of the first canonicalization, with and without the dense blocks, p. 56; here Table 9 | J1b, (ii) |
| Table 4.7 | Revised: first canonicalization and re-solve of the DPP problems, with standard deviations and significance, p. 60; here Table 5 | J1a, (iv), Q4 |
| Table A.1 | The 25 benchmark problems, p. 74; here Table 14 | (v) |
| Table A.2 | Dimensions of the benchmark problems, p. 76; here Table 15 | (v) |
| Table 4.5 | Ratio at seven densities, with and without the dense blocks, p. 58; here Table 11 | J1c, J1d |
| Table 4.6 | Ratio to the best `CVXPY` backend at five sizes, p. 59; here Table 12 | J1c, J1d |

## 3.1 First canonicalization (J1a, (iv))

Table 7 is the main result. `diffengine` is the fastest backend on 21 of the 25 problems, significantly on all 21, and its ratio to the best `CVXPY` backend is 0.44 in geometric mean. The table is sorted by that ratio, so the four slower problems are the last four rows; I explain each one in §3.4.

Table 7: Table 4.1 of the thesis (p. 53): `ignore_dpp` canonicalization time in milliseconds on all 25 problems, mean ± standard deviation over the runs. Bold denotes the fastest of the four backends on each problem. The ratio is the median time of `diffengine` over that of the best of `CVXPY`'s three backends. Every ratio is significant. The final row is the geometric mean of the ratios.

| Benchmark | CPP | SCIPY | COO | diffengine | Ratio |
|---|---|---|---|---|---|
| SVMWithL1Regularization | 2,786 ± 8 | 2,634 ± 8 | 2,742 ± 15 | **387 ± 1** | 0.15 |
| FactorCovarianceModel | 1,223 ± 5 | 1,164 ± 3 | 1,150 ± 4 | **177 ± 3** | 0.15 |
| Yitzhaki | 938 ± 1 | 982 ± 2 | 1,006 ± 3 | **145 ± 1** | 0.15 |
| SimpleQP | 2,891 ± 5 | 2,944 ± 6 | 2,929 ± 6 | **453 ± 1** | 0.16 |
| HuberRegression | 2,770 ± 6 | 2,850 ± 7 | 2,796 ± 7 | **495 ± 8** | 0.18 |
| LeastSquares | 2,410 ± 9 | 2,912 ± 7 | 2,939 ± 5 | **435 ± 2** | 0.18 |
| SimpleLP | 6,898 ± 38 | 6,303 ± 17 | 5,962 ± 14 | **1,121 ± 3** | 0.19 |
| SimpleScalarParametrizedLP | 1,260 ± 3 | 1,151 ± 2 | 1,087 ± 3 | **230 ± 1** | 0.21 |
| SimpleFullyParametrizedLP | 611 ± 1 | 555 ± 1 | 523 ± 1 | **115 ± 0.5** | 0.22 |
| SlowPruning | 1,792 ± 4 | 2,185 ± 4 | 2,147 ± 4 | **408 ± 1** | 0.23 |
| ParametrizedQP | 2,974 ± 29 | 3,030 ± 7 | 2,990 ± 10 | **755 ± 46** | 0.25 |
| Cajas | 3,885 ± 12 | 689 ± 5 | 693 ± 5 | **205 ± 1** | 0.30 |
| ConeMatrixStuffing | 2,517 ± 10 | 2,152 ± 8 | 3,537 ± 8 | **642 ± 4** | 0.30 |
| SmallMatrixStuffing | 1,683 ± 13 | 1,822 ± 8 | 2,508 ± 7 | **502 ± 2** | 0.30 |
| TvInpainting | 919 ± 1 | 913 ± 2 | 815 ± 2 | **257 ± 1** | 0.32 |
| ParamSmallMatrixStuffing | 63.3 ± 0.2 | 127 ± 1 | 113 ± 1 | **32.7 ± 0.1** | 0.52 |
| ParamConeMatrixStuffing | 42.8 ± 0.2 | 71.9 ± 0.3 | 88.7 ± 0.4 | **22.4 ± 0.4** | 0.52 |
| CVaR | 7,196 ± 15 | 6,503 ± 20 | 6,512 ± 23 | **3,421 ± 9** | 0.53 |
| SemidefiniteProgramming | 1,161 ± 3 | 632 ± 2 | 997 ± 4 | **405 ± 3** | 0.64 |
| QuantumHilbertMatrix | 1,062 ± 2 | 805 ± 4 | 788 ± 2 | **653 ± 3** | 0.83 |
| Murray | 1,276 ± 18 | 1,229 ± 7 | 1,224 ± 7 | **1,181 ± 1** | 0.96 |
| SDPSegfault1132 | 8,000 ± 11 | **1,663 ± 148** | 7,298 ± 110 | 2,147 ± 6 | 1.32 |
| OptimalAdvertising | 2,573 ± 3 | 598 ± 8 | **534 ± 1** | 901 ± 4 | 1.68 |
| ConvexPlasticity | 78.2 ± 0.7 | 72.0 ± 0.7 | **69.8 ± 1.2** | 507 ± 1 | 7.31 |
| UnconstrainedQP | **1,955 ± 3** | 1,980 ± 5 | 2,439 ± 2 | 22,937 ± 21 | 11.73 |
| **Geometric mean** | | | | | **0.44** |

## 3.2 Dense blocks (J1b, (ii), Q2)

Table 8 groups the problems by the effect of the dense blocks, and the groups follow the data.

**Dense blocks help.** On these twelve problems, the constant data is mostly nonzero, and turning the dense blocks off makes `diffengine` 1.2 to 3 times slower. On ConeMatrixStuffing and SmallMatrixStuffing, this data is the dense matrix of the objective $\mathbf{1}^\top \mathbf{A}\mathbf{x}$, which is why the density of their constraint matrix in Table 15 is low.

**Little effect.** On these ten problems, turning the dense blocks off changes the time by at most 7 %, and by at most 3 % on all but OptimalAdvertising. Murray, for instance, has a dense array of differences, but at a density below 5 %, where `diffengine` already stores a

Table 8: Table 4.4 of the thesis (p. 57): `ignore_dpp` canonicalization time of `diffengine` in milliseconds with and without the dense blocks, mean ± standard deviation over the runs, grouped by the effect of the dense blocks. The ratio is the median time without the dense blocks over that with them. Every ratio is significant except that of ParametrizedQP.

| Benchmark | `diffengine` | no dense blocks | Ratio |
|---|---|---|---|
| *Dense blocks help* | | | |
| SVMWithL1Regularization | 387 ± 1 | 745 ± 5 | 1.92 |
| FactorCovarianceModel | 177 ± 3 | 366 ± 6 | 2.06 |
| Yitzhaki | 145 ± 1 | 263 ± 1 | 1.82 |
| SimpleQP | 453 ± 1 | 914 ± 6 | 2.01 |
| HuberRegression | 495 ± 8 | 1,015 ± 3 | 2.06 |
| LeastSquares | 435 ± 2 | 906 ± 2 | 2.08 |
| SimpleLP | 1,121 ± 3 | 1,485 ± 5 | 1.32 |
| ConeMatrixStuffing | 642 ± 4 | 1,941 ± 7 | 3.03 |
| SmallMatrixStuffing | 502 ± 2 | 1,369 ± 6 | 2.73 |
| ParamSmallMatrixStuffing | 32.7 ± 0.1 | 40.4 ± 0.3 | 1.23 |
| ParamConeMatrixStuffing | 22.4 ± 0.4 | 27.4 ± 0.7 | 1.22 |
| SDPSegfault1132 | 2,147 ± 6 | 3,466 ± 15 | 1.61 |
| *Little effect* | | | |
| SimpleScalarParametrizedLP | 230 ± 1 | 225 ± 0.3 | 0.98 |
| SimpleFullyParametrizedLP | 115 ± 0.5 | 115 ± 0.4 | 1.00 |
| SlowPruning | 408 ± 1 | 417 ± 1 | 1.02 |
| ParametrizedQP | 755 ± 46 | 746 ± 34 | 1.00 |
| Cajas | 205 ± 1 | 209 ± 1 | 1.02 |
| TvInpainting | 257 ± 1 | 250 ± 1 | 0.97 |
| SemidefiniteProgramming | 405 ± 3 | 406 ± 1 | 1.00 |
| QuantumHilbertMatrix | 653 ± 3 | 671 ± 3 | 1.03 |
| Murray | 1,181 ± 1 | 1,179 ± 2 | 1.00 |
| OptimalAdvertising | 901 ± 4 | 834 ± 1 | 0.93 |
| *Dense blocks hurt* | | | |
| CVaR | 3,421 ± 9 | 2,419 ± 36 | 0.71 |
| ConvexPlasticity | 507 ± 1 | 97.1 ± 3.2 | 0.19 |
| UnconstrainedQP | 22,937 ± 21 | 500 ± 2 | 0.02 |

constant factor in the sparse format. These small differences are significant only because the runs vary so little.

**Dense blocks hurt.** On these three problems, a dense block is built from data that is mostly zeros, and turning the dense blocks off makes `diffengine` 1.4 to 46 times faster. The data is the scenario matrix of CVaR, a dense array with one nonzero in four; the diagonal of ConvexPlasticity, given as a dense array; and the chain of products of UnconstrainedQP, whose dense Jacobian comes from a sparse factor (§3.4).

The dense blocks speed up the problems with dense data, slow down the three that build

them from mostly zeros, and change little on the rest.

## 3.3 Peak memory (J1b, (ii), Q5)

Table 9: Table 4.3 of the thesis (p. 56): growth of the peak resident memory during the first canonicalization in megabytes on all 25 problems, mean over the runs. Bold denotes the least of the four backends. The last column is `diffengine` without the dense blocks.

| Benchmark | `CPP` | `SCIPY` | `COO` | `diffengine` | no dense blocks |
|---|---|---|---|---|---|
| SVMWithL1Regularization | **1,976** | 2,041 | 2,904 | 2,034 | 2,417 |
| FactorCovarianceModel | 1,129 | 1,047 | 1,334 | **760** | 999 |
| Yitzhaki | 449 | **437** | 504 | 476 | 572 |
| SimpleQP | 2,194 | 2,355 | 3,236 | **1,655** | 2,204 |
| HuberRegression | 2,223 | 2,246 | 3,121 | **1,413** | 1,866 |
| LeastSquares | 2,142 | 2,327 | 3,021 | **1,459** | 1,893 |
| SimpleLP | 5,353 | 4,961 | 5,027 | **4,792** | 5,057 |
| SimpleScalarParametrizedLP | 1,088 | 1,024 | 1,068 | **1,009** | 1,009 |
| SimpleFullyParametrizedLP | 517 | 494 | 526 | **492** | 492 |
| SlowPruning | **1,352** | 1,425 | 1,828 | 1,797 | 1,120 |
| ParametrizedQP | 2,035 | 2,174 | 2,998 | **1,775** | 1,775 |
| Cajas | 1,193 | **298** | 320 | 341 | 342 |
| ConeMatrixStuffing | 2,010 | 964 | 3,730 | **478** | 1,623 |
| SmallMatrixStuffing | 1,189 | 634 | 2,405 | **322** | 1,117 |
| TvInpainting | **705** | 858 | 994 | 1,048 | 1,048 |
| ParamSmallMatrixStuffing | 17.4 | 12.4 | 21.9 | **10.6** | 10.5 |
| ParamConeMatrixStuffing | 9.8 | 8.2 | 12.2 | **5.9** | 5.8 |
| CVaR | 4,022 | **2,804** | 3,587 | 10,654 | 2,115 |
| SemidefiniteProgramming | 1,317 | **350** | 640 | 1,138 | 1,138 |
| QuantumHilbertMatrix | 222 | **218** | 288 | 1,194 | 1,194 |
| Murray | 2,849 | **220** | 305 | 306 | 307 |
| SDPSegfault1132 | 7,536 | **3,460** | 15,628 | 10,413 | 12,885 |
| OptimalAdvertising | **173** | 182 | 197 | 6,066 | 945 |
| ConvexPlasticity | 82.5 | 82.5 | **82.3** | 1,329 | 25.5 |
| UnconstrainedQP | 1,427 | **1,058** | 2,105 | 1,754 | 644 |

Without the dense blocks, the peak memory falls most on the problems where the dense blocks also cost time: from 10.7 to 2.1 GB on CVaR, from 6.1 to 0.95 GB on OptimalAdvertising, from 1.3 GB to 26 MB on ConvexPlasticity and from 1.75 to 0.64 GB on UnconstrainedQP. It rises on the problems with dense data, where a dense block needs no index per entry: by 19 to 33 % from SVMWithL1Regularization to LeastSquares, and by a factor of about 3.4 on the two MatrixStuffing problems. In geometric mean, `diffengine` uses 0.83 times its memory with the dense blocks. On OptimalAdvertising, `diffengine` still

uses five times the memory of `CVXPY` without the dense blocks, which is the cost of the global layout (Q5).

## 3.4 The four slower problems ((vi), Q5)

Table 10 gives the share of each phase of the first canonicalization of `diffengine` on the four problems on which it is slower. On a typical problem, the time is spread over the phases; on each of the four, one or two phases take almost all of it, and the paragraphs below say why. Three of the four are addressed by changes to `CVXPY`, which the measurements of the thesis do not include; each fix is measured alone.

Table 10: Table 4.2 of the thesis (p. 54): phases of the first canonicalization of `diffengine` on the four problems on which it is slower, in percent of its time, and the median share over the 25 problems, mean over the runs. The phases are the conversion of the `CVXPY` expressions into the trees of the engine, the symbolic pass, the numeric pass, the assembly of $\mathbf{A}$ and $\mathbf{P}$ in CSC format, and the formatting of the rows by cone. The front end is the rest of the time, the reductions of `CVXPY` before the engine.

| | | share of the time (%) | | | | | |
|---|---|---|---|---|---|---|---|
| Benchmark | Time (s) | Front end | Convert | Symbolic | Numeric | Assemble | Format |
| UnconstrainedQP | 22.94 | 0 | 0 | 2 | 98 | 0 | 0 |
| ConvexPlasticity | 0.51 | 6 | 2 | 47 | 6 | 38 | 0 |
| OptimalAdvertising | 0.90 | 4 | 1 | 89 | 4 | 1 | 1 |
| SDPSegfault1132 | 2.15 | 0 | 0 | 49 | 47 | 3 | 0 |
| Median, all 25 | — | 4 | 9 | 19 | 14 | 12 | 12 |

**UnconstrainedQP.** The problem minimizes $\|\mathbf{F}^{\mathsf{H}}\mathbf{E}(\mathbf{z})\mathbf{F}\mathbf{x}-\mathbf{y}\|_2$, with $\mathbf{E}(\mathbf{z}) = \mathbf{I}_{14}\otimes\operatorname{diag}(\mathbf{z})$, over $\mathbf{z}\in\mathbb{C}^{18}$, where $\mathbf{F}$ is the $252\times252$ discrete Fourier transform matrix. Evaluated from the left, the product $(\mathbf{F}^{\mathsf{H}}\mathbf{E})\mathbf{F}$ has a dense Jacobian of $63{,}504\times36$ entries, which the numeric pass fills one entry at a time. Evaluated from $\mathbf{x}$, as the reassociation of matrix products excluded from the thesis measurements does, every intermediate is a vector, and the canonicalization takes 16 milliseconds instead of 22.9 seconds, with a peak memory of 53 megabytes instead of 1.75 gigabytes.

**OptimalAdvertising.** The problem maximizes $\sum_{i=1}^{250}\min(r_i\mathbf{p}_i^\top\mathbf{d}_i, b_i)$ subject to $\mathbf{D}\geq\mathbf{0}$, $\mathbf{D}^\top\mathbf{1}\leq\mathbf{t}$ and $\mathbf{D}\mathbf{1}\geq\mathbf{c}$, over $\mathbf{D}\in\mathbb{R}^{250\times1{,}000}$ with rows $\mathbf{d}_i^\top$. Each minimum term depends on one row of $\mathbf{D}$, but its Jacobian spans all 250,000 variables, and the symbolic pass visits every column of it, so its cost grows with the number of terms times the number of variables. The two aggregate constraints are written as products with a vector of ones, which become dense blocks. A change to `CVXPY` lowers them to sums, which reduces the canonicalization from 0.91 to 0.33 seconds, below the 0.53 seconds of the fastest `CVXPY` backend, and the peak memory from 6.1 to 1.4 gigabytes.

**SDPSegfault1132.** Three dense blocks of 49.5 million entries cancel in a sum with 1.01 million nonzeros, and `diffengine` carries the cancelled entries through both passes.

**ConvexPlasticity.** The diagonal $h\mathbf{I}$ of the quadratic objective is given as a dense array and becomes a dense 3,000 × 3,000 block of $\mathbf{P}$. A change to `CVXPY` stores $\mathbf{P}$ sparse below 5 % nonzeros, as is already done for a constant factor of a matrix product, which reduces the canonicalization from 509 to 203 milliseconds and the peak memory from 1.33 to 0.59 gigabytes. The problem remains 2.9 times slower than on the fastest `CVXPY` backend, and most of the remaining time is in the symbolic pass.

On each slower problem, one or two phases take almost all of the time, and three of the four improve with a change made before the data reaches the engine.

## 3.5 Density and size (J1c, J1d)

Table 11: Table 4.5 of the thesis (p. 58): ratio of the median canonicalization time of `diffengine` to that of the best `CVXPY` backend as the density $\rho$ of the data matrix decreases, with and without the dense blocks. Below $\rho = 0.05$, `diffengine` stores a constant factor in the sparse format. Every difference is significant.

| | density $\rho$ | | | | | | |
|---|---|---|---|---|---|---|---|
| Benchmark | 1 | 0.5 | 0.2 | 0.1 | 0.05 | 0.02 | 0.01 |
| *with dense blocks* | | | | | | | |
| SVMWithL1Regularization | 0.15 | 0.33 | 0.64 | 0.98 | 1.47 | 0.63 | 0.72 |
| FactorCovarianceModel | 0.15 | 0.33 | 0.57 | 0.77 | 1.01 | 0.60 | 0.65 |
| Yitzhaki | 0.15 | 0.29 | 0.53 | 0.75 | 0.99 | 0.67 | 0.73 |
| SimpleQP | 0.16 | 0.30 | 0.47 | 0.56 | 0.63 | 0.40 | 0.41 |
| HuberRegression | 0.18 | 0.34 | 0.67 | 1.03 | 1.78 | 1.74 | 2.54 |
| LeastSquares | 0.18 | 0.34 | 0.61 | 0.83 | 1.20 | 0.67 | 0.81 |
| SemidefiniteProgramming | 0.64 | 0.67 | 1.02 | 1.28 | 1.46 | 1.60 | 1.66 |
| Lasso | 0.15 | 0.33 | 0.59 | 0.84 | 0.58 | 0.75 | 0.86 |
| *without dense blocks* | | | | | | | |
| SVMWithL1Regularization | 0.28 | 0.32 | 0.36 | 0.41 | 0.48 | 0.62 | 0.71 |
| FactorCovarianceModel | 0.32 | 0.35 | 0.39 | 0.44 | 0.52 | 0.61 | 0.66 |
| Yitzhaki | 0.28 | 0.30 | 0.36 | 0.46 | 0.56 | 0.66 | 0.71 |
| SimpleQP | 0.32 | 0.34 | 0.39 | 0.44 | 0.47 | 0.51 | 0.52 |
| HuberRegression | 0.37 | 0.36 | 0.46 | 0.62 | 0.96 | 1.72 | 2.52 |
| LeastSquares | 0.38 | 0.36 | 0.40 | 0.43 | 0.53 | 0.66 | 0.79 |
| SemidefiniteProgramming | 0.64 | 0.68 | 1.02 | 1.28 | 1.46 | 1.61 | 1.67 |
| Lasso | 0.31 | 0.34 | 0.37 | 0.45 | 0.54 | 0.69 | 0.79 |

As the density of the data falls (Table 11), the ratio with the dense blocks rises until about $\rho = 0.05$ on the problems that use them, because a dense block keeps storing and multiplying every entry while the `CVXPY` backends do less work. Below $\rho = 0.05$, `diffengine` stores the constant in the sparse format, and the ratio drops to about that without the dense blocks. Without them, the ratio rises slowly as the density falls. The two variants are about equal

at $\rho = 0.5$, and the variant without them is faster from $\rho = 0.2$ to 0.05. The dense blocks pay off only when about half or more of the entries are nonzero.

Table 12: Table 4.6 of the thesis (p. 59): ratio of the median canonicalization time of `diffengine` to that of the best `CVXPY` backend as the size of each problem is scaled by $s$, in the order of Table 4.1. "cap" denotes a size at which `diffengine` exceeded the 20 GB memory limit while a `CVXPY` backend finished, "all cap" one at which every backend exceeded it, and a dash a size the problem does not have. Every difference is significant.

| | size factor $s$ | | | | |
|---|---|---|---|---|---|
| Benchmark | 1/4 | 1/2 | 1 | 2 | 4 |
| SVMWithL1Regularization | 0.16 | 0.16 | 0.15 | 0.15 | 0.16 |
| FactorCovarianceModel | 0.17 | 0.16 | 0.15 | 0.18 | 0.22 |
| Yitzhaki | 0.14 | 0.16 | 0.15 | 0.14 | 0.14 |
| SimpleQP | 0.15 | 0.15 | 0.16 | 0.16 | 0.16 |
| HuberRegression | 0.16 | 0.19 | 0.18 | 0.17 | 0.17 |
| LeastSquares | 0.16 | 0.16 | 0.18 | 0.17 | 0.18 |
| SimpleLP | 0.20 | 0.20 | 0.19 | 0.18 | all cap |
| SimpleScalarParametrizedLP | 0.22 | 0.22 | 0.21 | 0.20 | 0.19 |
| SimpleFullyParametrizedLP | 0.24 | 0.22 | 0.22 | 0.21 | 0.20 |
| SlowPruning | 0.23 | 0.22 | 0.23 | 0.22 | 0.21 |
| ParametrizedQP | 0.17 | 0.23 | 0.25 | 0.23 | 0.22 |
| Cajas | 0.29 | 0.28 | 0.30 | 0.26 | 0.26 |
| ConeMatrixStuffing | 0.31 | 0.31 | 0.30 | 0.29 | 0.31 |
| SmallMatrixStuffing | 0.42 | 0.30 | 0.30 | 0.27 | 0.27 |
| TvInpainting | 0.33 | 0.33 | 0.32 | 0.31 | 0.29 |
| ParamSmallMatrixStuffing | 0.52 | 0.52 | 0.52 | 0.51 | 0.43 |
| ParamConeMatrixStuffing | 0.51 | 0.52 | 0.52 | 0.52 | 0.52 |
| CVaR | 0.59 | 0.56 | 0.53 | cap | all cap |
| SemidefiniteProgramming | 0.58 | 0.61 | 0.64 | 0.64 | 0.68 |
| QuantumHilbertMatrix | — | 1.17 | 0.83 | all cap | — |
| Murray | 1.06 | 1.10 | 0.96 | 1.00 | cap |
| SDPSegfault1132 | 1.47 | 1.39 | 1.32 | cap | all cap |
| OptimalAdvertising | 0.80 | 1.11 | 1.68 | 3.01 | cap |
| ConvexPlasticity | 3.32 | 5.40 | 7.31 | 8.82 | cap |
| UnconstrainedQP | 5.73 | 7.59 | 11.73 | 16.78 | 22.64 |
| Lasso | 0.16 | 0.15 | 0.15 | 0.14 | 0.15 |

As each problem is scaled from $s = 1/4$ to $s = 4$ (Table 12), the ratio stays within 0.15 of its value at the original size on 21 of the 25 problems, at every size where `diffengine` finishes, so the speedup of Table 7 does not come from the particular sizes of the suite. On three of the others, slower problems whose cost grows faster than their data, the ratio grows for the reasons of §3.4: on OptimalAdvertising the number of terms and the number of variables grow together, on ConvexPlasticity the dense block of $\mathbf{P}$ grows with the square of the number of points, and on UnconstrainedQP the dense Jacobian of the chain grows with the size of the Fourier matrix.

`diffengine` exceeds the 20 GB memory limit at five sizes where a `CVXPY` backend finishes. Without the dense blocks, three of them finish, at ratios of 0.31 (CVaR, $s = 2$), 1.57 (ConvexPlasticity, $s = 4$) and 3.18 (OptimalAdvertising, $s = 4$). The other two exceed the

limit with and without the dense blocks: SDPSegfault1132 already needs 10 to 13 GB at its original size (Table 9), and at $s = 4$ on Murray, `CPP` exceeds it as well. At the four sizes marked "all cap", every backend exceeds it. The ratio does not depend on the sizes of the suite, except on three of the slower problems, and the dense blocks cause three of the five sizes at which only `diffengine` runs out of memory.

## 3.6 Re-solve against a full canonicalization (Q4)

Table 13: Not in the thesis. Re-solve time in milliseconds on the seven DPP problems, mean ± standard deviation over 30 re-solves, each with new parameter values: the cached tensor of `CVXPY`, a full canonicalization on the `ignore_dpp` path, both with the fastest backend, and the cached re-solve of `diffengine`.

| | `CVXPY` | | `diffengine` |
|---|---|---|---|
| Benchmark | tensor | recompile | cached |
| ParamSmallMatrixStuffing | 1.84 ± 0.01 | 57.6 ± 0.2 | 0.438 ± 0.007 |
| ParamConeMatrixStuffing | 0.958 ± 0.005 | 37.7 ± 0.2 | 0.265 ± 0.004 |
| SimpleFullyParametrizedLP | 46.7 ± 0.1 | 519 ± 1 | 34.6 ± 0.8 |
| SimpleScalarParametrizedLP | 62.8 ± 0.3 | 1,078 ± 2 | 70.4 ± 5.9 |
| ParametrizedQP | 299 ± 8 | 2,868 ± 14 | 349 ± 16 |
| SVMWithL1Regularization | 38.3 ± 0.1 | 2,570 ± 3 | 125 ± 2 |
| FactorCovarianceModel | 20.5 ± 0.1 | 1,138 ± 3 | 74.2 ± 0.8 |

Table 13 compares both caches with what `CVXPY` does without one, canonicalizing again. A re-solve from the cached tensor is 10 to 67 times faster than a full canonicalization, and a re-solve of `diffengine` 8 to 142 times faster. Either cache removes most of the cost of a re-solve, and the gap between the two, at most 3.6 times (Table 5), is small next to it.

## 3.7 The benchmark problems ((v))

Tables 14 and 15 are Appendix A of the thesis, the description and the dimensions of each problem, for reference in the tables above.

Table 14: Table A.1 of the thesis (p. 74): the 25 problems of the benchmark suite.

| Benchmark | Description |
|---|---|
| SVMWithL1Regularization | Hinge-loss classifier with an $\ell_1$ penalty, 25,000 samples and 500 features. The penalty weight is a parameter. |
| FactorCovarianceModel | Factor-model portfolio selection with 25,000 assets and 250 factors. The risk aversion and the leverage limit are parameters. |

| Benchmark | Description |
|---|---|
| Yitzhaki | Gini mean-difference portfolio LP over a dense matrix of all 44,850 pairwise return differences of 50 assets. |
| SimpleQP | QP with a dense $1{,}600 \times 1{,}600$ quadratic form, 8,000 inequality and 20 equality constraints. |
| HuberRegression | Huber regression with 3,000 features and 4,500 samples, 12 % of the labels flipped. |
| LeastSquares | Dense least squares, $\|\mathbf{Ax} - \mathbf{b}\|_2^2$ with $\mathbf{A} \in \mathbb{R}^{7000 \times 2000}$. |
| SimpleLP | Box-constrained LP with $10^7$ variables and a linear objective. |
| SimpleScalarParametrizedLP | Box-constrained LP with $2 \times 10^6$ variables whose cost vector is scaled by a scalar parameter. |
| SimpleFullyParametrizedLP | Box-constrained LP with $10^6$ variables whose cost vector is a parameter. |
| SlowPruning | Matrix least squares with a $100 \times 20$ variable and dense data, from `CVXPY` issue 1668. |
| ParametrizedQP | Box-constrained least squares whose $6{,}000 \times 2{,}400$ matrix and right-hand side are parameters. |
| Cajas | Gini portfolio LP in ordered-weighted-average form, with 800 vector constraints added one by one. |
| ConeMatrixStuffing | The LP of SmallMatrixStuffing at size 5,000. |
| SmallMatrixStuffing | LP with 4,000 variables and 4,000 scalar constraints, each a separate constraint object. |
| TvInpainting | Total-variation inpainting of a $512 \times 512$ colour image, with about 30 % of the pixels known. |
| ParamSmallMatrixStuffing | The LP of SmallMatrixStuffing at size 300, with its matrix, constraint coefficients and right-hand side as parameters. |
| ParamConeMatrixStuffing | The same parametric LP at size 200. |
| SemidefiniteProgramming | Standard-form SDP with a $200 \times 200$ variable and 120 dense equality constraints. |
| CVaR | Conditional value-at-risk LP over a dense matrix of 131,072 scenarios and 768 variables. |
| QuantumHilbertMatrix | SDP with partial traces and partial transposes of a $64 \times 64$ positive semidefinite variable, with sparse constants. |
| Murray | The Yitzhaki LP written with a dense $244{,}650 \times 700$ matrix of pairwise differences. |
| SDPSegfault1132 | Fits a distance matrix through a $99 \times 99$ positive semidefinite Gram matrix and dense constants, from `CVXPY` issue 1132. |
| OptimalAdvertising | Allocation of 250 advertisers over 1,000 time slots, with one minimum term per advertiser. |
| ConvexPlasticity | One step of a convex plasticity model at 3,000 points, with a von Mises yield criterion and three parameters. |

| Benchmark | Description |
|---|---|
| UnconstrainedQP | Fits a complex signal through a $252 \times 252$ discrete Fourier transform matrix and a Kronecker product with a diagonal of the variable, from `CVXPY` issue 2205. Despite its name, a second-order cone problem. |

Table 15: Table A.2 of the thesis (p. 76): dimensions of the problem data of the 25 benchmark problems.

| Benchmark | $m$ | $n$ | nnz($\mathbf{A}$) | density (%) | nnz($\mathbf{P}$) | param. size | DPP |
|---|---|---|---|---|---|---|---|
| SVMWithL1Regularization | 51 k | 26 k | 12.6 M | 0.95 | — | 1 | yes |
| FactorCovarianceModel | 75.3 k | 75.2 k | 6.45 M | 0.11 | 87.5 k | 2 | yes |
| Yitzhaki | 89.8 k | 44.9 k | 4.57 M | 0.11 | — | — | — |
| SimpleQP | 8,020 | 1,600 | 12.8 M | 99.8 | 2.56 M | — | — |
| HuberRegression | 13.5 k | 16.5 k | 13.5 M | 6.1 | 4,500 | — | — |
| LeastSquares | 7,000 | 9,000 | 14 M | 22.2 | 7,000 | — | — |
| SimpleLP | 20 M | 10 M | 20 M | $1.0 \times 10^{-5}$ | — | — | — |
| SimpleScalarParametrizedLP | 4 M | 2 M | 4 M | $5.0 \times 10^{-5}$ | — | 1 | yes |
| SimpleFullyParametrizedLP | 2 M | 1 M | 2 M | $1.0 \times 10^{-4}$ | — | 1 M | yes |
| SlowPruning | 400 k | 402 k | 8.4 M | $5.2 \times 10^{-3}$ | 400 k | — | — |
| ParametrizedQP | 10.8 k | 8,400 | 14.4 M | 15.9 | 6,000 | 14.4 M | yes |
| Cajas | 641 k | 2,450 | 1.96 M | 0.12 | — | — | — |
| ConeMatrixStuffing | 5,000 | 5,000 | 5,000 | 0.020 | — | — | — |
| SmallMatrixStuffing | 4,000 | 4,000 | 4,000 | 0.025 | — | — | — |
| TvInpainting | 2.61 M | 1.05 M | 3.63 M | $1.3 \times 10^{-4}$ | — | — | — |
| ParamSmallMatrixStuffing | 300 | 300 | 300 | 0.33 | — | 90.5 k | yes |
| ParamConeMatrixStuffing | 200 | 200 | 200 | 0.50 | — | 40.3 k | yes |
| SemidefiniteProgramming | 20.2 k | 20.1 k | 2.43 M | 0.60 | — | — | — |
| CVaR | 264 k | 132 k | 25.7 M | 0.074 | — | — | — |
| QuantumHilbertMatrix | 6,176 | 6,176 | 81.1 k | 0.21 | 4,096 | — | — |
| Murray | 490 k | 245 k | 1.5 M | $1.2 \times 10^{-3}$ | — | — | — |
| SDPSegfault1132 | 15 k | 4,951 | 1.01 M | 1.4 | — | — | — |
| OptimalAdvertising | 252 k | 250 k | 1 M | $1.6 \times 10^{-3}$ | — | — | — |
| ConvexPlasticity | 36 k | 33 k | 72 k | $6.1 \times 10^{-3}$ | 9.05 M | 27 k | no |
| UnconstrainedQP | 1,009 | 289 | 18.6 k | 6.4 | — | — | — |

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